Shan Yang

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EXPERIENCE

Amazon Prime Video – Senior Applied Scientist Tech Lead | June 2025 - present
Leading Prime Video GenAI Live Action Studio * Lora/ControlNet fine tuning for character and scene consistency and pose controllability, laid foundation of AI Studio Roadmap * New temporal diffusion model training (1B from scratch and open source model finetuning) * Leading foundational video generation research * 4 Publications Pending

Amazon Search - Senior Applied Scientist Tech Lead | Sept 2021 - June 2025
Led Amazon product search multi-modal personalization * Led and launched Amazon Video Search on Amazon product search page * Designed Amazon Video Search Ranker (model architecture and feature design) and Product Video Understanding Models (based on CLIP model) * Designed Amazon Video Search System * Led Multi-modal LLM for Video Captioning and Question Answering (new Schema-of-Thought COT for VQA method) * Led full-stack development & A/B Testing * Amazon Patent

Google Research - Senior Software Engineer | Jun 2017 - Sept 2021
Multimodal Video Content Creation * Multimodal Dance Generation *
3D Vision Research: Monocular Video Pose Understanding * Owner and major contributor of the 2D/3D Reconstruction ML Model Development * Created end-to-end monocular video analyzer for biological research * Google Patent

Google Research, Mountain View — Research Intern | Summer 2016 (3 months)
3D Vision Research: Monocular Video 3D Human Pose Understanding * 3D Hair Reconstruction
Samsung Research America, San Jose —Software Engineer Intern | Summer 2014 (3 months) and Summer 2013 (3 months)
Advanced Processor Lab * Graphics hardware test infrastructure development

EDUCATION

University of North Carolina at Chapel Hill, NC —Ph.D. Computer Science | 2012 - 2017
Shanghai Jiao Tong University, China — B.E. Software Engineering | 2008 - 2012

INTERNAL REPORTS

Schema-of-Thought for Robust Video Question Answering, Xijun Wang, **Shan Yang**
visual chain-of-thought rooted from human perception psychology, +8% vs. ChatGPT-4o, +6% vs. Claude3.5

Linear Layer for Test-Time Reasoning, Peihao Wang, **Shan Yang**
new linear layer adapter to LLM to enhance reasoning capability, +10% vs. Llama3 on math,

ClipDreamer: Towards Faithful and Visually-Grounded Product Video Generation, Zun Wang, **Shan Yang**
new conditional generation to ground generation on input video, +8% on motion correctness

Temporal Diffusion Model, Peihao Wang, **Shan Yang**
new temporal diffusion SDE for temporal consistency improvement

PUBLICATIONS

ViLA: Efficient video-language alignment for video question answering
Xijun Wang, Junbang Liang, Chun-Kai Wang, Kenan Deng, Yu Lou, Ming C Lin, **Shan Yang**, European Conference on Computer Vision (ECCV), 2024

ICAR: Image-based Complementary Auto Reasoning
Xijun Wang, Anqi Liang, Junbang Liang, Ming Lin, Yu Lou, **Shan Yang**, AAAI 2023

MeSa: Masked, Geometric, and Supervised Pre-training for Monocular Depth Estimation
Muhammad Osama Khan, Junbang Liang, Chun-Kai Wang, **Shan Yang**, Yu Lou, NeurIPS 2023 Workshop SSLTheoryPractice

Attention Bottlenecks for Multimodal Fusion
Arsha Nagrani, **Shan Yang**, Anurag Arnab, Aren Jansen, Cordelia Schmid, Chen Sun, Neural Information Processing Systems (NIPS), 2021

Optical Mouse: 3D Mouse Pose From Single-view Video
Shan Yang*, Bo Hu*, David A. Ross, Avneesh Sud, Yi Liu, Graham Ruby, Bryan Seybold, Conference on Computer Vision and Pattern Recognition (CVPR), 2021 CV4Animal Workshop

Learn to Dance with AIST++: Music Conditioned 3D Dance Generation
Shan Yang*, Ruilong Li*, David A. Ross, Angjoo Kanazawa, International Conference on Computer Vision (ICCV),

2021

Physics-Inspired Garment Recovery from a Single-View Image
Shan Yang, Zherong Pan, Tanya Amert, Ke Wang, Licheng Yu, Tamara Berg, Ming C. Lin, ACM Transaction on Graphics (TOG), 2018

Learning-based Cloth Material Recovery from Video
Shan Yang, Junbang Liang, Ming C. Lin, International Conference on Computer Visio (ICCV), 2017

Modeling Context in Referring Expressions
Lichens Yu, Patric Poirson, **Shan Yang**, Alex Berg, Tamara Berg, European Conference on Computer Vision (ECCV), spotlight, 2016

Classification of Prostate Cancer Grades and T-Stages based on Tissue Elasticity Using Medical Image Analysis
Shan Yang, Vladimir Jojic, Jun Lian, Ronald Chen, Hongtu Zhu and Ming C. Lin, International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2016

Bayesian Estimation of Non-Rigid Mechanical Parameters Using Temporal Sequences of Deformation Samples
Shan Yang, Ming C. Lin, IEEE International Conference on Robotics and Automation (ICRA), 2016

Material Cloning: Acquiring Elasticity Parameters from Images
Shan Yang, Ming C. Lin, IEEE Transactions on Visualization and Graphics (TVCG), 2016

Simultaneous Estimation of Elasticity for Multiple Deformable Bodies
Shan Yang, Ming C. Lin, Journal Computer Animation and Virtual Worlds (CAVW), 2015